TinyLLM lab — own 5.6M model, Oct 24, 2026

Kaggle Tesla T4, Karpathy llama2.c, TinyStories. Trained from scratch, not fine-tuned.

In one minute

- Loss fell from 10.39 (near random) to 1.75 train / 1.76 val in 10,000 steps.
- Grammar arrived by 5.6M params. Holding onto an entity did not.
- More params bought entity preservation first, then story structure. Temperature only changed how the model picked, not what it knew.

Loss notes

Config: dim 144, 4 layers, 4 heads, max seq len 128, batch 32, grad accum 2, 8,192 tokens per step, learning rate 1e-3 with warmup 200, dtype float32.

- Step 0: train loss 10.3901, val loss 10.3903
- Step 10,000: train loss 1.7507, val loss 1.7617
- Last step loss 1.8334 is single-batch noise, not the eval number.
- Learning rate decayed to 0.0 at the end, as scheduled.
- Total tokens seen: about 82M (8,192 x 10,000).
- Train/val gap: 0.011, so no overfitting inside TinyStories.
- Checkpoint: `out-tiny/model.bin`, saved locally as `tinyllm-own-5m-10k-2026-10-24-model.bin`, with `tokenizer.bin`.

Random guessing over a 32,000-token vocabulary gives loss about 10.37. Loss 1.75 means about 17% average probability on the correct next token, versus 0.003% at random.

Samples — own model

Prompt for all three: "One day, Lily met a Shoggoth"

Temperature 0.8 — 163.2 tok/s

> One day, Lily met a Shoggoth. It was big and had a shiny shape. Lily wanted to play with it, so she took turns pulling it. She was having fun, but then she started to feel sad. "Ouch! That's a big bug!" said Lily. "I have an idea!" "I will push you, Lily!" said the bug. "I will help you!" So, they started to move and make a noise. Suddenly, the bug landed on the table and landed on the floor. "Oh no! You did

Note: story-like, but Shoggoth becomes "bug" after the first sentence, dialogue roles swap, repetition, and the output cuts off at the token limit. No arc.

Temperature 0.2 — 152.5 tok/s

> One day, Lily met a Shoggoth in the park. The grumpy cat was very nice and liked to play with Lily. They played together and had lots of fun. But then, Lily saw a big dog. The dog was not nice. It wanted to play with Lily. Lily said, "No, this is my toy. It is not nice." The dog did not listen. He wanted to play with the dog. Lily and the dog played with the dog. They had fun. But then, something unexpected happened. The dog started to bark! Lily

Note: most grammatical, most repetitive. Shoggoth is lost to a cat, then a dog loop. Low temperature did not preserve the entity, it just made the safest picks repeat.

Temperature 1.5 — 133.5 tok/s

> One day, Lily met a Shoggoth in her friendly yard.ain loved Lily a while because that soap dry her house, through anymore. other clothes saw that sometimes, difficult seems ugly, and we get thetruck tervation though easily notice her where you Amy walked behind. She clapped and said, "Your toyss arrive now." Everyone together for them, Buddy was proud of new body that she watched out for fires. His backpacks la distant ago made everything easier she could apple! Me Gre Gold physics quietly striking tele PC pulled backwards towardsrem explos raised up its moves easier

Note: grammar collapses after the first clause into word salad, with merged tokens and unintroduced characters. At this size, high temperature adds noise, not creativity.

Scale comparison, same Shoggoth prompt

Model	Speed	Entity	Arc
Own 5.6M	133-163 tok/s	Lost after 1 sentence, to bug / cat / dog	None
Pretrained 15M	50.5 tok/s	Lost to "shampoo", circular definition, repetition loop	None
Pretrained 42M	17.8 tok/s	Kept as a creature, name drifts (Shoggoth / Shoggg / Shogglow)	Yes, weak to strong to dancing

Pretrained outputs above are summaries from the Oct 24 comparison, not full pasted text. Own-model outputs above are verbatim.

Files in this folder

- `tinyllm-own-5m-10k-2026-10-24-model.bin`
- `tokenizer.bin` (required for local `./run`, keep this exact name)
- This summary